

MINI PROJECT REPORT
ON
“Attainment Calculation Software”
SUBMITTED IN PARTIAL FULLFILLMENT OF THE REQUIREMENTS
OF DEGREE OF
BACHELOR OF ENGINEERING
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[2023-24]



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2023-24
Certificate

This is to certify that the Mini Project-1A entitled “**Attainment calculation Software**” is a bonafide work of **Zeeshan Shaikh, Suryoday kumar, Sahil Shaikh, Hrutik Parbalkar** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of “**Undergraduate**” in “**Computer Engineering**”.

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Mini Project-1A Report Approval

This project report entitled “**Attainment Calculation Software**” submitted by “**Zeeshan Shaikh, Suryoday kumar, Sahil Shaikh, Hrutik Parbalkar**” is approved for the degree of **Bachelor of Engineering in Computer Engineering**.

Examiners

1. _____

2. _____

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Zeeshan Shaikh

Suryoday kumar

Sahil Shaikh

Hrutik Parbalkar

Date:

Place:

Abstract

An efficient method for assessing course outcomes (CO) and program outcomes (PO) in engineering program accreditation. It emphasizes the importance of using direct and indirect assessment tools, such as internal tests, assignments, and final exams, to evaluate student performance accurately. The paper provides comprehensive guidelines for calculating attainment by selecting appropriate assessment methods to measure COs and POs. Each CO is designed to align with specific program outcomes. Correlation levels (1, 2, or 3) are employed to gauge the strength of CO-PO mapping, with higher levels indicating a stronger alignment. Target levels (low, medium, high) are set for each CO, and attainment is evaluated using both direct and indirect methods. The results are presented in a comparative table, enabling faculty to identify gaps and take actions to enhance overall student proficiency.

KEYWORDS

Course Outcome, Education, Engineering Education, Outcome Based Education, Program Outcomes, Excel.

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Chapter 1

Introduction

Introduction

In the realm of higher education, assessing the attainment of learning outcomes is paramount to ensure the effectiveness of educational programs. College attainment calculators play a crucial role in this process by quantifying the achievement of specific learning objectives, known as Course Outcomes (COs), through direct and indirect measures. These calculators traditionally rely on predefined formats and formulas, limiting their adaptability to various datasets and educational contexts.

Our project aims to address this limitation by introducing customization options that empower users to tailor the calculator to their specific needs. By allowing users to input row numbers, column numbers, sheet names, and calculation formulas, our enhanced calculator can seamlessly integrate with any Excel sheet without the need for formatting adjustments. This not only improves the usability of the calculator but also enhances its flexibility and applicability across different educational settings.

This report provides a detailed overview of our enhanced college attainment calculator, highlighting its features, advantages, and the methodology behind its development. We will also present the results of using the calculator with various datasets, demonstrating its effectiveness in analyzing both direct and indirect attainment. Through this project, we aim to contribute to the advancement of educational assessment tools and provide educators with a more versatile and user-friendly solution for evaluating learning outcomes.

Objectives:

The objective of the proposed system is to implement an automated attainment calculation system to calculate course outcome and Course outcome attainment of a particular course. The user interface will allow users to enter CO-PO mapping. Target level, direct and indirect assessment data. After accepting all the data, the system generates a CO. The system also generates CO attainment in the observation sheet.

The proposed System can be divided into mainly two parts, namely inserting values and attainment calculation. User interface is designed with staff and admin credentials. Admin has access to upload, edit and delete student lists for the current academic year. The staff has the option of inserting data such as CO-PO matrix, direct and indirect assessment values.

Data Collection Optimization: Implement efficient methods for collecting data from course exit surveys and internal assessments, ensuring accuracy and timeliness.

Weighted Response Analysis: Assign appropriate weights to survey responses based on their relevance to course objectives and use these weights to calculate direct attainment scores accurately.

Internal Assessment Integration: Establish a seamless process for collecting and aggregating internal assessment and coursework data from various departments and academic years.

Attainment Calculation Algorithm: Design and implement a robust algorithm for calculating both direct and indirect attainment scores, accounting for variations in data sources.

Visualization Interface: Develop an intuitive and user-friendly interface for visualizing attainment data, allowing stakeholders to easily interpret and analyze the results.

Integration of Educational Improvement Plans: Ensure that the College Attainment Calculator can seamlessly integrate with the institution's educational improvement plans and initiatives.

Chapter 2

Literature Survey

2.1 Basic Terminologies

- **User friendly interface:** The system encompasses an interface which is easy to interact, intuitively designed and accessible to users.
- **Excel sheet access:** System is able to access the excel sheet which is on the user's device, so it can take inputs for calculations.
- **Data visualization:** It encompasses the representation of data in bar graphs hence the educators are able to take further valuable decisions.
- **Single step computations:** System envelopes the calculations without any challenges and struggle.

Direct Attainment:

Definition: Direct attainment refers to the measurement of student learning outcomes based on their responses to course exit surveys or assessments that directly evaluate their performance against specific learning objectives.

Explanation: It assesses the immediate impact of a course on a student's knowledge and skills. This measurement directly reflects how well the course has achieved its intended learning outcomes.

Indirect Attainment:

Definition: Indirect attainment is the assessment of student learning outcomes through data collected from internal assessments, coursework, and other forms of evaluation that indirectly reflect the alignment with educational goals.

Explanation: Indirect attainment provides insights into how well the overall curriculum and teaching methods support the achievement of educational objectives. It takes into account the broader educational experience.

Course Exit Survey:

Definition: A course exit survey is a questionnaire or assessment completed by students at the end of a course to gather feedback on their learning experience, the course content, and the effectiveness of teaching methods.

Explanation: These surveys help in measuring direct attainment by collecting student feedback about their understanding of course objectives and whether the course met their expectations.

Internal Assessment:

Definition: Internal assessment involves evaluating students' performance through various methods such as exams, assignments, projects, and other coursework conducted within the institution.

Explanation: Internal assessments are a primary source for calculating indirect attainment, as they reflect how well students perform in their coursework in relation to learning outcomes.

Rubrics:

Definition: Rubrics are structured scoring guides that outline specific criteria and performance levels used to assess the quality of student work, assignments, or projects.

Explanation: Rubrics are often used in the evaluation of internal assessments to ensure consistency and objectivity in assessing how well students have met course objectives.

Data-Driven Decision-Making:

Definition: Data-driven decision-making is a process in which decisions are made based on analysis of relevant data and information, rather than relying solely on intuition or anecdotal evidence.

Explanation: In the context of the College Attainment Calculator, it means using the attainment data collected to make informed decisions regarding curriculum improvement and educational quality enhancement.

Dashboard:

Definition: A dashboard is a visual interface that displays key data, metrics, and performance indicators in a centralized and easily accessible manner, often in the form of charts, graphs, and tables.

Explanation: Dashboards in the context of this project provide a visual overview of attainment data, making it easier for stakeholders to understand and interpret the results at a glance.

2.2 Existing system

- **Manual data calculations:** Existing system is based on manual data calculations. So, in this system data calculations system has added to ease in calculations.
- **User Interface Challenges:** Outdated user interfaces can be cumbersome and unintuitive, making it difficult for users to navigate and use the software effectively.
- **Lack of Mobile Accessibility:** Many existing systems may not be optimized for mobile devices, limiting accessibility for both educators and students.
- **Inflexibility:** Some systems lack flexibility and customization options, preventing educators from tailoring the software to meet their specific needs.
- **Manual Data Collection** In the existing system, data collection from course exit surveys, internal assessments, and coursework is primarily done manually.
- **Course exit survey responses** are collected on paper or through simple online forms, which can be time-consuming and error-prone.

Internal assessment results and coursework data are often managed through spreadsheets or other basic tools.

- **Lack of Integration** Data from various sources, such as surveys and internal assessments, are typically stored in different locations and may not be well-integrated.
This lack of integration makes it challenging to perform comprehensive analysis and to assess attainment in a holistic manner.
- **Data Silos** Data is often siloed within individual departments or courses, limiting the ability to perform cross-departmental or cross-course analysis.
This makes it difficult to identify trends or areas for improvement at the institution level.
- **Time-Consuming Analysis** Analyzing data, especially for indirect attainment, can be a time-consuming and manual process. Faculty and administrators may spend significant time compiling and analyzing data, which could be used for more strategic activities.
- **Limited Visualization** The existing system may lack effective tools for visualizing data, making it challenging for stakeholders to understand and interpret the data easily. Decision-makers may rely on raw data and tables without graphical representations.
- **Lack of Automation** Many processes related to data collection, analysis, and reporting are manual and lack automation. This can lead to errors, inconsistencies, and delays in assessing student attainment.
- **Difficulty in Long-Term Analysis** Long-term analysis of student attainment trends may be limited due to the absence of historical data and tools to track changes over academic years.

2.3 Problem Statement

- In education, tracking and assessing student attainment is vital for evaluating curriculum effectiveness and teaching quality. Our project focuses on creating an attainment calculator that processes Excel data to calculate direct and indirect attainment scores for college courses and programs.

- The college currently lacks a systematic and automated system for direct and indirect attainment score calculation. And Manual data collection by faculty and administrators is time-consuming and error-prone.
- Manual Data Collection: The current system relies heavily on manual data collection methods for both course exit surveys and internal assessments. This process is time-consuming, error-prone, and lacks automation.
- Data Silos: Data from various sources, such as surveys, assessments, and coursework, are often stored in department-specific silos. This makes it challenging to perform institution-wide analysis and identify trends.
- Inconsistent Analysis: The analysis of student attainment data, particularly for indirect attainment, is inconsistent and time-consuming. There is a lack of standardized processes and tools for effective analysis.
- Limited Visualization: Stakeholders often struggle to interpret raw data, as there is a lack of effective data visualization tools. This hinders the ability to quickly understand and act upon the data.
- Inability to Monitor Trends: There is limited capacity to monitor long-term trends in student attainment over multiple academic years, which is essential for strategic planning and quality enhancement.
- Impact of the Problem: The inability to comprehensively assess student learning outcomes can result in a misalignment between courses and educational objectives. Inefficient data collection and analysis processes can lead to delays in identifying areas for improvement, impacting the overall quality of education. The lack of data-driven decision-making capabilities limits the institution's ability to make informed changes to curriculum and teaching methods. This can affect student satisfaction, retention, and the institution's reputation.

Chapter 3

Requirements Gathering

3.1 Software and Hardware Requirements

Here we will discuss everything we will need in order to execute. Below we list the necessary hardware and software requirements.

1. Software Requirements:

Visual Studio: You can use these compilers to compile the code.

HTML: It is a markup language used to create web pages and other types of documents that can be displayed in a web browser.

CSS: It is a style sheet language used for describing the presentation of a document written in HTML (Hypertext Markup Language) or XML (Extensible Markup Language)

Java script: JavaScript is a widely-used programming language for creating interactive and dynamic web applications. It enables client-side scripting to enhance user experiences on websites.

2. Hardware Requirements:

- Processor: INTEL CORE i5
- Memory: 4 GB Ram

Chapter 4

Plan of Project

4.1 System Architecture

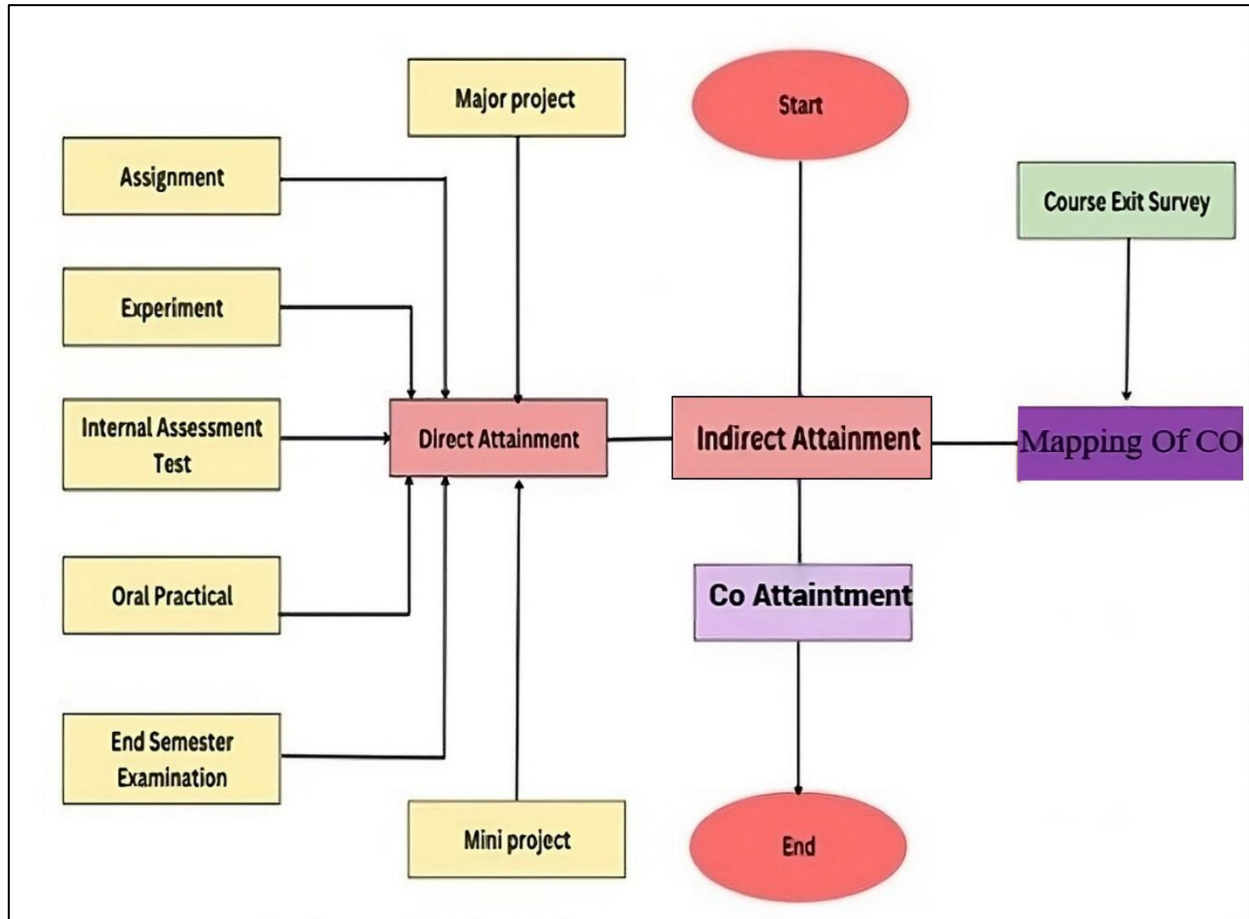


Figure 4.1: System Architecture

4.2 Methodology

The methodology used for the development of the Attainment Calculating Software for the college project involved the following steps:

Requirements gathering: The first step involved understanding the requirements of the project and identifying the features and functionalities that the project should have. This was done by consulting with the college authorities, faculty members, and students, and by reviewing existing college websites and resources.

Design and planning: Based on the requirements identified in the first step, we developed a design plan for the Attainment Calculator, which included the user interface, the features and functionalities, and the programming languages and technologies to be used.

Implementation: The implementation phase involved coding the chatbot using HTML, CSS, JavaScript and other relevant programming languages and APIs.

Testing: We conducted various tests on the software to ensure that it was user friendly, reliable, and capable of handling a variety of user inputs and scenarios. We also checked the bot for security vulnerabilities and implemented appropriate measures to protect user data.

Data Collection

Course Exit Surveys: Develop an online survey platform for students to complete at the end of each course. This survey should include questions that align with course objectives and provide insights into students' perception of their learning experiences.

Internal Assessments and Coursework: Create a system for collecting internal assessment and coursework results, which should be standardized and easily accessible for analysis.

Data Processing:

Implement data processing algorithms to validate, clean, and structure the collected data. This includes handling missing responses, data entry errors, and ensuring consistency.

Direct Attainment Calculation:

Design an algorithm to calculate direct attainment based on course exit survey data. This algorithm should involve:

Assigning weights to survey questions based on their relevance to course objectives.

Aggregating weighted responses to calculate direct attainment scores for each course.

Indirect Attainment Calculation:

Develop an algorithm for calculating indirect attainment based on internal assessment and coursework data. This algorithm should include the use of rubrics to assess the alignment of coursework with learning outcomes. Aggregating student performance scores across courses and academic years.

Attainment Visualization:

Create a user-friendly dashboard and data visualization tools to present direct and indirect attainment data in a comprehensible manner. This may involve charts, graphs, and tables to provide insights into the data.

Cross-Departmental Analysis:

Enable the system to perform cross-departmental and cross-course analysis by integrating data from various departments and academic years. This allows for identifying trends and areas for improvement across the institution.

Automation and Integration:

Develop features that automate data collection, processing, and attainment calculation. Ensure integration with existing systems or databases to streamline the process.

Long-Term Trend Analysis:

Incorporate tools to track and analyze long-term trends in student attainment data over multiple academic years. This facilitates strategic planning for continuous improvement.

Testing and Quality Assurance:

Conduct thorough testing to identify and rectify any issues in the software. Ensure data accuracy, security, and system performance.

Training and Implementation:

Provide training to relevant staff and faculty members on how to use the College Attainment Calculator. Support the implementation of the tool in the educational institution.

Data Security and Privacy:

Implement robust security measures to protect sensitive student data, ensuring compliance with data protection regulations.

Continuous Improvement and Maintenance:

Establish a process for ongoing system maintenance and improvement, including periodic updates and enhancements based on user feedback and changing educational needs.

Chapter 5

Result Analysis

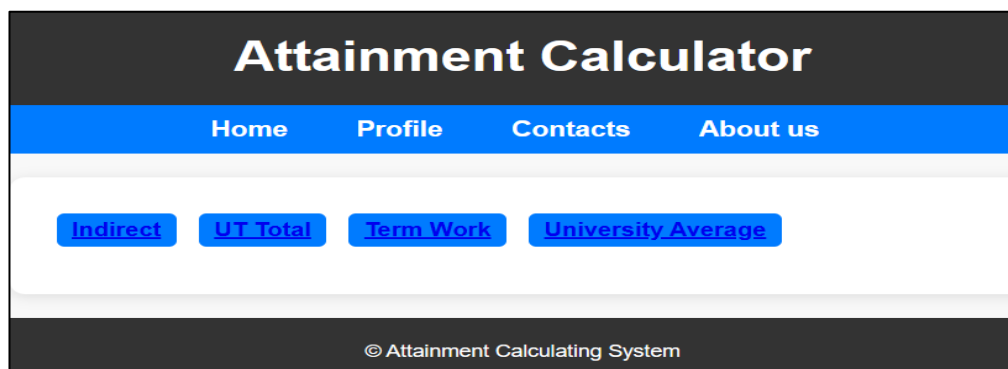


Fig 5.1 Attainment Calculation

INDIRECT ASSESSMENT REPORT							
COURSE EXIT SURVEY STATISTICS							
Total no. of student present for exist survey:				102			
Course objectives	Acceptance level					%	Out of 20
	5	4	3	2	1		
CO1	46	51	5	0	0	88.04	17.61
CO2	52	38	12	0	0	87.84	17.57
CO3	48	45	8	1	0	87.45	17.49
CO4	46	42	13	1	0	86.08	17.22
CO5	46	43	13	0	0	86.47	17.29
CO6	45	45	12	0	0	86.47	17.29
						Avg :	17.41
Choose File miniproject_excel.xlsx							
View Bar Graph							

Fig 5.2 Indirect Assessment Report

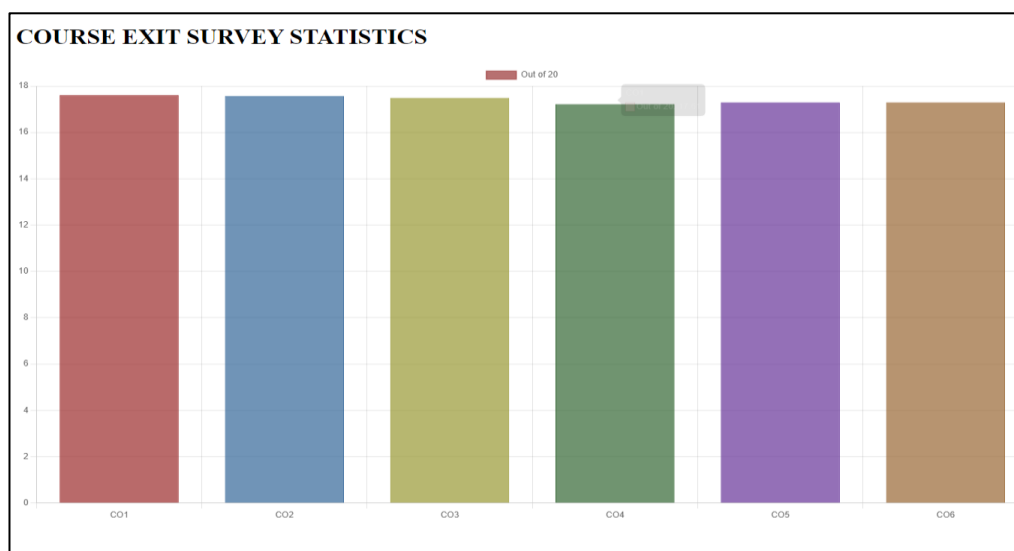


Fig 5.3 Course Exit Survey Statistics

Direct Table of Term Work				
Distribution of marks for CO	Ass 1	Ass 2	Exp	Total
CO1	1.42	0.46	2.25	4.13
CO2	0.83	1.04	3.00	4.87
CO3	2.75	0.58	3.75	7.08
CO4	0.00	1.25	1.50	2.75
CO5	0.00	0.63	2.25	2.88
CO6	0.00	1.04	2.25	3.29

Fig 5.4 Direct Table of Term work

Distribution of Marks for CO	20
CO1	4
CO2	3
CO3	4
CO4	2
CO5	3
CO6	4
Total	20

Distribution of Marks for CO	Out of 80
CO1	8
CO2	13
CO3	20
CO4	17
CO5	7
CO6	18
Total	80

Fig 5.5 Distribution of marks for CO

Max Marks	20	25	25	80
Distribution of marks for CO1	4	4	5	7
Distribution of marks for CO2	3	5	5	13
Distribution of marks for CO3	4	7	6	19
Distribution of marks for CO4	2	3	7	17
Distribution of marks for CO5	3	3	3	6
Distribution of marks for CO6	4	3	3	18
Total	20	20	25	80

% Attainment of CO level				
	IA	TW	Oral	ESE
CO1	18.10	12.87	15.35	7
CO2	13.57	16.08	15.35	7
CO3	18.10	22.52	18.43	18
CO4	9.05	9.65	9.21	16
CO5	13.57	9.65	9.21	6
CO6	18.10	9.65	9.21	17

Fig 5.6 Direct Attainment for CO

UT Attainment

Desired Columns:

D,E,F,G,H,I

Start Row For Sums:

148

End Row For Sums:

169

Start Row For Ones:

10

End Row For Ones:

140

Choose File miniproject_excel.xlsx

Fig 5.7 Ut Attainment

Sheet Number:

Columns to Extract (comma-separated):

Start Row:

End Row:

Total Number of Students:

miniproject_excel.xlsx

Sheet Index:

Start Column:

End Column:

Start Row:

End Row:

miniproject_excel.xlsx

UNI PAPER	CO1	CO2	CO3	CO4	CO5	CO6
out of 100	9	16	24	21	8	22
out of 80	7.20	12.80	19.20	16.80	6.40	17.60
Final round out of 80	7	13	19	17	6	18

Fig 5.8 University Attainment

Chapter 6

Conclusion

6.1 Conclusion

The proposed system is able to accept values and calculate the attainment. This system will be used by various colleges & universities for CO attainment calculation. At the same time, this system can also be used to redesign course outcomes and program outcomes to stay updated with the industry requirements. This system is secured with user and admin credentials. Currently, this system only generates attainment for Course outcomes. In the future, AI & ML techniques can be used to display the action that needs to be taken if a Course outcome is not achieved.

6.2 References

Akash Rajak, Ajay Kumar Shrivastava, Divya Prakash Shrivastava, "Automating Outcome Based Education for the Attainment of Course and Program Outcomes," IEEE, 2020.

Prof. Vinay Kumar Chandna, "Course Outcome Assessment and Improvement on Weak Student," IEEE, 2015

Acknowledgement

It is a privilege for us to have been associated with **Prof. Ekta Ukey**, our guide, during this project work. We have been greatly benefited by her valuable suggestions and ideas. It is with great pleasure that we express our deep sense of gratitude to them for their valuable guidance, constant encouragement and patience throughout this work. We are also indebted to our guide for extending the help to academic literature. We express our gratitude to **Prof. Dhanashri Sakhare** (Project Coordinator), **Prof. Rohini Bhosale** (Head of Department of Computer Engineering), **Dr. J. W. Bakal** (Principal) for their constant encouragement, cooperation and support. We take this opportunity to thank all our classmates for their company during the course work and for useful discussion we had with them. We would be failing in our duties if we do not make a mention of our family members, including our parents for providing moral support, without which this work would not have been completed.

Thanking You,

Zeeshan Shaikh
Suryoday Kumar
Sahil Shaikh
Hrutik Parbalkar